

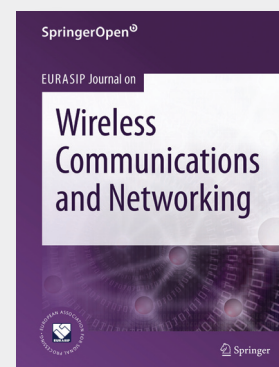


EURASIP Journal on Wireless Communications and Networking

Special Issue on
Research methodologies for performance
evaluation of wireless network solutions

This special issue originates from the FIRE projects CREW and EULER and solicits papers related to performance evaluation methods for future internet or communication research, for instance contributions from the FIRE community in Europe and the GENI initiative in the US. FIRE refers to the objective 'Future Internet Research and Experimentation (FIRE)' under the 'Cooperation' programme theme Information and Communication Technologies (ICT) of the 7th Framework Programme (2007 – 2013) of the European Community for research, technological development and demonstration activities (FP7). Global Environment for Network Innovations (GENI) is a program supported by the National Science Foundation (NSF). Both FIRE and GENI programs target at-scale experimentation on shared, heterogeneous, highly instrumented infrastructure in view of promoting innovations in network science and engineering.

This special issue publishes key contributions on objective and credible methods for the performance evaluation of wireless network solutions.



Call for
Papers

Potential topics include, but are not limited to:

- Verification of analytical models and simulation environments against real-life
- Proven practices for validation of wireless networking solutions through simulation or experimentation
- Benchmarking methods enabling fair comparison between competing solutions or subsequent developments
- Unambiguous definitions of performance evaluation metrics and according methods for measurement and calculation
- Tools specialized for realistic traffic generation and the creation of repeatable background environment properties (for example home, office, factory) in simulators or experimental settings
- Methods and techniques to generate repeatable, reproducible, and reliable measurements in experimental homogeneous/heterogeneous settings
- Analysis of control tools for configuring and driving experiments, e.g. management and control tools to conduct experiments with ease and efficiency, monitor the experiment conditions, and run-time modifying the experiment parameters
- Distributed measurement techniques and tools for test-beds/experimental facilities including small to large-scale test-beds, federated test-beds, homogeneous/heterogeneous test-beds, etc
- Run-time integration of modeling and simulation tools into test-beds/experimental facilities
- Data-mining, interpretation, knowledge acquisition from simulation or experimental measurements
- Statistical analysis of experimental data including multivariate analysis (e.g., analysis of variance, regression analysis, principal component analysis and variants, discriminant analysis and variants), analysis time series (e.g. frequency-domain/time-domain methods, parametric and non-parametric methods, univariate/multivariate), statistical inference (e.g., frequentist, Bayesian)

Submission Instructions:

Before submission authors should carefully read over the Instructions for Authors, which are located at <http://jwcn.eurasipjournals.com/authors/instructions>. Prospective authors should submit an electronic copy of their complete manuscript through the SpringerOpen submission system at <http://jwcn.eurasipjournals.com/manuscript> according to the submission schedule. They should choose the correct Special Issue in the “sections” box upon submitting. In addition, they should specify the manuscript as a submission to the “Special Issue on Research methodologies for performance evaluation of wireless network solutions” in the cover letter. All submissions will undergo initial screening by the guest editors for fit to the theme of the Special Issue and prospects for successfully negotiating the review process.

Submission Schedule

Manuscripts due:
January 31, 2014

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